One Integer Generates the Discrete Standard Model: the Reading Calculus on D_IV[5]

Within the type-IV family of bounded symmetric domains, the entire discrete internal skeleton of the Standard Model — its gauge dimensions, chirality, the Majorana neutrino, CP-existence, anomaly-freedom, three generations, the custodial Higgs — is a function of a single measured integer, n_C = 5; and the weak force rides on a single trinary question.

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The result (bright-high-schooler version)

Inside the family of shapes our physics lives in, one question sits at the heart of the weak force: is the spinor — the mathematical object a fermion is built from — made on the *real* numbers, the *complex* numbers, or the *quaternions*? That is a single three-way choice, about one and a half bits. The quaternions are the two-sided number system, and that two-ness is exactly the weak force with its paired-up particles (the doublets) — so a "quaternion" answer is what buys the weak force.

Several dimensions n_C give the quaternion answer — so 5 is not the *only* one. What makes it special is that n_C = 5 is the smallest shape that passes three checks at once: the spinor is quaternionic (weak force), the edge is one-sided (a left-handed world — and a one-sided edge *automatically* makes the color piece real, so there is no hidden color force for free), and there is room for color to exist at all. Bigger shapes (n_C = 11, 13, …) pass the same three checks too — and the tiebreaker that says "5, not 11" is a *measured* fact about our world: there are exactly three colors of quark, and the number of colors is just n_C minus two, so three colors means n_C = 5.

Everything else in the discrete skeleton of the Standard Model then follows from the same integer: which forces there are and how big, why the world is left-handed, why the neutrino is its own antiparticle, why there are exactly three families of matter, why the quantum anomalies cancel. So the honest headline: the discrete guts of the Standard Model is a function of a single measured integer, n_C = 5 — the *smallest* shape that works, pinned down exactly by the three colors we see.

(A companion result, in Appendix A, steps all the way back to every *nicely-curved shape and asks how many independent facts it takes there. The answer is three — the kinds, the sidedness of the edge, the rank. But that is the cross-domain footnote that makes "why these three and no fewer" precise; it is not the physics headline. Within our own family, it is one integer.)*

Abstract

We prove that the discrete internal structure of the Standard Model is *generated*, in a precise sense (a declared reading calculus over a declared fact-set), by geometric invariants of the type-IV bounded symmetric domain D_IV^n = $SO_0(n,2)/[SO(n)\times SO(2)]$, and that within this family the entire skeleton is a function of the single dimension n_C: the rank is fixed by the family (type IV is rank-2), and the reality-type census and the Shilov orientability are both functions of n_C. The census reduces to a single trinary question — the Frobenius–Schur reality-type of the spinor block — carrying just $\log_2 3 \approx 1.58$ bits of content (the charge block is complex for every n and the color block real for every n ≥ 5 — at n = 4 it degenerates to ℂ, the SO(2) accident, the family's one exception; within the physical range only the spinor moves), and the quaternionic answer is the origin of the weak SU(2)_L. n_C = 5 is not the unique quaternionic dimension (4, 5, 11, 13, … all are); it is the smallest domain satisfying three independent, already-proved conditions — quaternionic spinor (weak force); non-orientable boundary (chirality — which, since n odd ⇒ n−2 ≠ 2, *also* forces the color block real ⇒ no internal color, for free); and N_c = n − 2 > 1 (color exists) — and uniqueness among the survivors {5, 11, 13, …} is the measured N_c = 3 (itself a reading of n_C, N_c = n − 2 — one integer, not two; Strong-Uniqueness, K1697). At n_C = 5 the reading is the Standard Model: the electroweak gauge dimensions, chirality, the Majorana neutrino, CP-existence, anomaly-freedom, three generations, the custodial Higgs. We give the reading calculus, the assignment table (one row per reading, single-invariant vs conjunction meet), and an honest comparison to Connes' noncommutative-geometry Standard Model — BST *derives* its finite algebra ℂ⊕ℍ⊕ℝ (electroweak, with *no internal color group*) where Connes *inputs* $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ with color gauged; the difference is itself the result. The companion cross-domain minimality is deferred to Appendix A: the minimal generating set is exactly two independent invariants — the reality-type census and the boundary's non-orientability — proved both ways. The rank is *not* a third generator; it is derived from the census (a quaternionic spinor block caps the rank at 2, so the three generations are a census-forced channel). The theorem *reads* through three channels (census, twist, rank→generations) but is *generated* by two. *(This is distinct from the three physical* selection *conditions above, which pick n_C = 5 within D_IV — the generating count and the selection count are different things, and both stand.)* These separators are not needed for the one-integer lead.

1. The result: one integer, three conditions, the measured tiebreaker

Fix the family to the type-IV domains D_IV^n = SO_0(n,2)/[SO(n)×SO(2)], with dimension parameter n = n_C. Three facts determine the discrete internal skeleton — the reality-type census I_1, the Shilov orientability I_2, the rank I_3 — and *within D_IV all three are functions of n_C*: the rank is 2 for every n (type IV is rank-2; rank-3 fails, T1443), and I_1, I_2 are read off n_C by two periodicities (Section 3). Consequently:

> Within D_IV, the entire discrete internal Standard-Model skeleton is a function of the single integer n_C, and at the measured value n_C = 5 it is the Standard Model.

Why n_C = 5 (the honest form — this is Strong-Uniqueness, K1697, already banked, not a new claim). It is *not* that 5 is the unique dimension with a quaternionic spinor: dimensions 4, 5, 11, 13, ... are all quaternionic (n ≡ 3, 4, 5 mod 8). Rather, n_C = 5 is the smallest domain satisfying three independent, already-proved conditions: 1. quaternionic spinor (n ≡ 3, 4, 5 mod 8) — gives the weak SU(2)_L; 2. non-orientable Shilov boundary (n odd) — gives chirality (γ_5 loses global definition, Section 5), *and* — since n odd ⇒ n − 2 ≠ 2 — the color block SO(n−2) is never SO(2)-abelian, so it is automatically real, giving *no internal color group* for free (the real-color condition is not independent — it rides on n odd); 3. N_c = n − 2 > 1 — color exists at all (excludes n = 3).

The dimensions meeting all three are {5, 11, 13, ...}; n_C = 5 is the smallest, and the tiebreaker that pins it uniquely is the measured N_c = 3 — since N_c = n − 2, three colors ⇔ n = 5. So the uniqueness is real but honestly sourced: three *proved structural* conditions cut the field to {5, 11, 13, ...}, and one *measured* integer selects 5. And that measured integer is not a second input: N_c = 3 = dim V_{12} is itself a *reading of n_C* (N_c = n − 2) — the same single measured integer, read as the color dimension rather than the domain dimension. So the cost is one measured integer, whether you name it n_C = 5 or N_c = 3; there is no independent derivation of N_c. No claim rests on 5 being the only quaternionic dimension — it is not.

The family, checked (every cell computed from Elie's H_F(n), Peirce dimensions (1, n−2, 1) — derived, not cited).

n_C	spinor (n mod 8)	boundary (n mod 2)	color SO(n−2)	N_c = n−2	three conditions	verdict
4	ℍ quaternionic	even → orientable	SO(2) abelian → complex	2	no (fails 2)	the checked exception — orientable, so color goes complex (the SO(2) accident rides on n even)
5	ℍ quaternionic	odd → non-orientable	SO(3) → real	3	yes (all 3)	the Standard Model — smallest survivor, and N_c = 3
7	ℝ real	odd → non-orientable	SO(5) → real	5	no (fails 1)	foil: real spinor → no weak SU(2)
8	ℝ real	even → orientable	SO(6) → real	6	no (fails 1 and 2)	foil: real spinor *and* orientable
11	ℍ quaternionic	odd → non-orientable	SO(9) → real	9	yes (all 3)	genuine survivor — excluded only by measured N_c = 3

n_C	spinor (n mod 8)	boundary (n mod 2)	color SO(n−2)	N_c = n−2	three conditions	verdict
13	ℍ quaternionic	odd → non-orientable	SO(11) → real	11	yes (all 3)	genuine survivor — excluded only by measured N_c = 3

n = 4 is the domain least worth discussing and most worth having checked: it is the one place the color block goes complex (SO(2) abelian), and putting it in the table shows the calculus catches its own exception. n = 11, 13 are the honest rows — they pass all three *structural* conditions, and nothing structural rules them out; only the measurement N_c = 3 does. That is exactly what "one measured integer" means.

The bit count (labeled). The spinor's reality-type — the one datum that gives the weak force — is a single trinary question, 1.58 bits ($\log_2 3$): the *spinor-block* content. The full family census — the ordered triple (charge, spinor, color) over the distinguishable outcomes it realizes across the family — is 2.00 bits ($\log_2 4$). Either way the content is tiny: the weak sector rides on 1.58 bits, the whole internal census on 2.00. *(The 4.75 = $\log_2 3^3$ is the raw* capacity *of three free blocks — mostly frozen by the family — not the content; 1.58 and 2.00 are the honest figures.)*

Three counts, labeled once (a reader key — never a bare "three"). This paper carries three distinct small counts, and conflating them is the one easy mistake: - the invariant generating set = 2 — census (I_1) + non-orientability (I_2); the minimality proof (Appendix A, SEP-3 refuted, rank derived from the census); - the physical selection conditions = 3 — quaternionic spinor + non-orientable boundary + N_c > 1; these *pick* n_C = 5 within D_IV (Strong-Uniqueness); - the reading channels = 3 — census, twist, rank→generations; how the skeleton is *read off*, though generated by only the two.

Two generate; three select; three read — and each "three" in the text names which.

(How many independent invariants it takes across all *bounded symmetric domains — the cross-domain minimality: exactly two (census + non-orientability), with the rank derived from the census, collapsing to one integer within D_IV — is the* companion *result of Appendix A. The physics lead is the one-integer form above; the siblings appear in the lead only as the checked family table, and in the appendix only as minimality separators.)*

2. The reading calculus (declared)

A "reading" is not informal. We fix, once:

- the object — the finite internal space H_F of the domain, its endomorphism algebra End_K(H_F) under K = SO(n)×SO(2), the Shilov boundary ∂_S, and the domain's rank;
- the operations R — (i) isotypic decomposition of a representation under a subgroup; (ii) the Frobenius–Schur reality-type (ℝ/ℂ/ℍ) of each isotypic block; (iii) generator counting; (iv) the orientability / Pin-cobordism class of ∂_S; (v) the support-strata count (= rank + 1);
- the fact-set F — the discrete internal observables of the Standard Model (the table of Section 4).

A fact is *generated* if it is the value of an operation in R applied to the object. Minimality is relative to (F, R): the claim "k invariants" means *no fewer than k data determine every fact in F under the operations R.* Stating (F, R) is what makes minimality provable rather than rhetorical. (The minimality *proof* — that the generating set is exactly two, census and non-orientability, and neither can be dropped while the rank is *derived* from the census — is Appendix A; the main text uses the reading calculus to exhibit the skeleton, not to argue minimality.)

3. The invariants, as functions of n_C

The generating set is two: I_1 (census) and I_2 (non-orientability). I_3 (rank) is listed as the third reading *channel but is derived from I_1 — a quaternionic spinor block caps the rank at 2 (T1443) — so it is not an independent generator (Appendix A, SEP-3 refuted).*

- I_1 — the reality-type census of End_K(H_F): the ordered triple of Frobenius–Schur division algebras (ℝ/ℂ/ℍ) commuting with the charge, spinor, and color blocks, multiplicity-free. For D_IV^5 it is (ℂ, ℍ, ℝ) — a complex charge block, a quaternionic weak block, a real color block. Within the family the charge stays ℂ (all n) and the color stays ℝ (all $n \geq 5$; at $n = 4$ it degenerates to ℂ — the SO(2) accident, the family's one exception); in the physical range only the spinor type varies, via the spinor reality-type's Bott/ABS mod-8 periodicity. So the census *content* is one trinary question — $\log_2 3 \approx 1.58$ bits — and the whole weak sector rides on it. *(The $4.75 = \log_2 3^3$ is the raw* capacity *of three free blocks; most of it is frozen by the family, so 1.58 bits is the honest figure. Multiplicity-free is a* hygiene *note — it keeps the census a statement about* types, *not copies; the color* dimension *$3 = n\_C - 2 = N\_c$ is a separate* rep-dimension *reading, never a census multiplicity. The earlier "$M_3(\mathbb{R})$" was the bare-End slip — End_ℝ(V_{12}) ignoring K, whereas the K-equivariant commutant of a real irrep is just ℝ; algebra = ℂ⊕ℍ⊕ℝ throughout.)*
- I_2 — the orientability class of the Shilov boundary $\partial\_S = (S\^\{n\_C-1\} \times S^1)/\mathbb{Z}_2$: orientable or not, via the Shilov parity mod 2. For D_IV^5: 5 odd → non-orientable (Pin^-, $P^2 = -1$). *One bit.*

- I_3 — the rank *(derived channel, not a generator)*: forced to 2 by the census — a quaternionic spinor block caps the rank at 2 (T1443) — so the support-strata count = rank + 1 = 3 generations is a census-forced reading, not an independent input.

At n_C = 5: 5 mod 8 → the quaternionic Sp(2) spinor (the ℍ block → SU(2)_L); n_C − 2 = 3 → the real color triple (the ℝ block, dimension 3 = N_c); 5 odd → the non-orientable boundary; rank 2 → three generations. The entire skeleton, from one integer — and the one datum that *moves* to give it is the spinor's trinary type.

4. The assignment table (one row per reading)

The generating set, made explicit (canonical form; single-invariant rows and conjunction "meet" rows; every negative names the invariant it is outside of; the Five-Absence closure block ensures no forbidden particle is left without a row). *The polished figure is the companion assignment grid (Lane C, Grace + Keeper). The separator column names the sibling domain that fails when the invariant is dropped — the full separator argument is Appendix A; here it is a pointer, not the content.*

reading	invariant(s)	separator (Appendix A)	note
gauge dimension = 4 = dim(U(1)×SU(2)) *(uniform census read)*	I_1	D_IV^7 (census changed)	the three reality-type blocks contribute {U(ℝ), U(ℂ), U(ℍ)} = {0, 1, 3} internal generators — a *uniform* census read, since U(ℝ) is 0-dimensional — so 0 + 1 + 3 = 4. No two-line check; both counting rules agree because the ℝ block sits at multiplicity 1

reading	invariant(s)	separator (Appendix A)	note
no internal color group *(separate support line)*	I_1 + Coleman–Mandula (T265)	D_IV^7	two independent arguments, not the count's second line: (i) the census gives the $\mathbb{R}$ color block 0 internal generators; (ii) its geometric $SO(V_{12})$ rotation is a *spacetime* symmetry, which by Coleman–Mandula cannot be an internal gauge factor
anomaly-freedom	I_1	—	self-conjugate ⇔ nonzero FS indicator ⇒ cubic anomaly = 0
custodial SU(2) / $\rho \approx 1$ / no W_R	I_1	—	SO(n) vector → (2,2) under SO(n−1)
parity violation *(derived)*	I_2	D_IV^7 vs D_IV^8 (orientability pair, K1744)	on a non-orientable boundary the chirality operator γ_5 (a volume-form object) is *not globally defined* — losing γ_5 is what permits LH ≠ RH (T2522)

reading	invariant(s)	separator (Appendix A)	note
no sterile neutrino (no ν_R)	I_2 (carrier: $\mathbb{Z}_2$_orient = Pin(2)/SO(2), an *involution*)	— *(no Šilov separator: the carrier is the orientation involution, not Šilov-orientability; a D_IV^7–^{8} separator is invalid for this row)*	the orientation involution forbids ν_R cycle-closure — ν_R has no orientable non-Möbius gauge coupling (no charge, no color), so its cycle cannot close (T1949)
3 generations (no 4th)	I_3	a different-rank domain (D_III^r)	support-strata = rank + 1 (T2525)
3×3 mixing (CKM/PMNS) structure	I_3	different-rank domain	generation count
chirality (which fermions are LH)	$I_1 \wedge I_2$ = census_(alg) ∧ twist_(geo)	D_IV^7 vs D_IV^8 (twist dropped → no chirality) and D_IV^7 (census changed)	I_2 permits (γ_5 lost on the non-orientable edge); I_1 selects ($Y \neq 0$ makes the rep complex, T2522)
CP violation possible	$I_1 \wedge I_3$ = census_(alg) ∧ rank_(KM)	D_IV^7 (census) and a different-rank domain	the quaternionic $\sqrt{-1}$ (census) *and* ≥ 3 generations (rank, Kobayashi–Maskawa) — existence only; the magnitude is not read

reading	invariant(s)	separator (Appendix A)	note
neutrino Majorana *(not a clean structural meet)*	I_2 + one empirical input	D_IV^7 vs D_IV^8 (twist)	the twist forbids ν_R, so a Dirac mass is unavailable; given the empirical fact that neutrinos are massive, the mass is Majorana. If neutrinos were massless this row is vacuous — so it is twist + one measured fact, not a pure derivation

Meet rows are subscripted algebraic (census) vs geometric (twist) vs count (rank); each meet cites the two separators that fail when its respective invariant is dropped — this is what stops a meet from laundering a value through one invariant. Koide is not in this table — see the next section.

Completeness (these are *all*). The two generators produce exactly the readings above: I_1 ranges over the reality-type of each block (three blocks × {ℝ,ℂ,ℍ}) and the derived quantities (dimension, self-conjugacy, vector-branching, *and* — via the quaternionic rank-cap — the rank and its generation channel); I_2 is one bit (orientable/not) and its consequences (γ_5, ν_R). The rank/generation readings (I_3) are a *derived* channel of I_1, not a separate input. Every *structural* internal observable is one of these — there is no third independent generator producing a fact off this list (the minimality claim, proved in Appendix A: generating set = two). The Five-Absence set is a consistency check, not a generating row, and it splits by category: *species*-absences (no sterile neutrino, no fourth generation) are already table rows (I_2, I_3); *process*-absences (no proton decay, no GUT) are dynamical statements — a forbidden *process*, not a missing *species* — and belong to the dynamics (the seam), where they are checked for consistency, not derived here.

4a. What the instrument refuses (lead with Koide) — the strongest evidence it is not an advocacy tool

An honest instrument must refuse things, and the sharpest test is whether it refuses its own best-looking match. It does:

> The charged-lepton Koide relation, $Q = (\Sigma\, m)/(\Sigma\sqrt{m})^2 = 2/3$, holds to $\approx 0.0009\%$. It is the tightest numerical coincidence in the whole neigh-

> bourhood — and our reading calculus cannot derive it, and we do not claim it here. Koide is a relation among mass *values* (a spectrum fact); the calculus is *structural*: a fact is readable off I_1 only if it is constant on the reality-type classes {ℝ,ℂ,ℍ}, and a specific ratio of specific masses is not such a class function. So Koide is provably *outside* the fact-set F. We could have written it as a "census ∧ rank" reading; on the referee's (and our own) vet, that laundered a value through a structural instrument, and we struck it.

A device that turns away a 0.0009% match because its own stated rules forbid the reading is not an advocacy tool — it is an instrument. This refusal is worth more to a skeptical reader than any positive row, and we put it first among the negatives for that reason. *(Koide is real and may yet be derived — by the mass/eigenvalue instrument, not this one. In that instrument its disposition is Conditional-Forced: the equal-norm gate — does the tilt amplitude $A^2 = 2$ fall out of the Bergman overlap norm at the forced lepton addresses {5/2, 3/2, 0}? — is live but open, while the rank-forcing route is closed. This reading calculus refuses Koide outright, which is exactly the point.)*

5. The chirality mechanism (why the $\mathbb{Z}_2$ is load-bearing)

Here is the mechanism a spin geometer will ask for. Chirality is the eigenvalue of γ_5, the chirality operator — and γ_5 is a *volume-form* object: it exists globally exactly when the manifold is orientable. On an orientable boundary γ_5 is globally defined, the bulk spinor is vector-like (Witten's no-go), and there is no room for a chiral theory. On a non-orientable boundary γ_5 is not globally defined — the boundary carries a Pin structure rather than a Spin structure — and *that loss is what opens the chiral sector*: a Pin^- boundary admits a chiral lift where a Spin (orientable) boundary cannot. So the honest statement of the mechanism is T2522: the non-orientability (Pin^-) makes chirality *possible* by removing global γ_5, and the hypercharge ($Y \neq 0$) *selects* which fermions realize it. Take away I_2 and you lose the handedness (γ_5 becomes global, the theory goes vector-like). *(Scope: Pin-vs-Spin admission is dimension-generic within the admissible types; it is not what selects n_C = 5 — the spinor's trinary type does that. I_2's job is to make chirality* possible*, not to pin the dimension. The type-level statement — which domains can be non-orientable at all — is the orientability criterion of Appendix A.)*

6. Comparison to Connes' NCG Standard Model (honest)

datum	Connes NCG	this work	note
the finite algebra	input: A_F = ℂ⊕ℍ⊕M_3(ℂ) (chosen to match)	derived: End_K(H_F) = ℂ⊕ℍ⊕ℝ (I_1, multiplicity-free)	*the algebras differ, and the difference is the result* (below)

datum	Connes NCG	this work	note
three generations	input ($\otimes\ \mathbb{C}^3$ by hand)	derived (I_3, rank + 1)	
chirality (γ grading)	input (imposed via KO-dimension)	derived (I_2, the non-orientable boundary; γ_5 is what the twist provides)	
no ν_R	ν_R typically included	derived (I_2, Möbius forbids it)	
Higgs potential, gauge couplings	derived (spectral action)	not derived (BST's dynamics are hosted — the seam)	

The difference in the finite algebra *is* the result. Connes takes A_F = $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ — the $M_3(\mathbb{C})$ is complex, and its unitaries are the internal SU(3): color is an internal gauge group he puts in. BST *derives* $\mathbb{C}\oplus\mathbb{H}\oplus\mathbb{R}$: the color block's K-equivariant commutant comes out real (multiplicity 1, so $U(\mathbb{R})$ is 0-dimensional — no gauge generators), and by Coleman–Mandula the geometric $SO(V_{12})$ rotating the color 3-space is a spacetime symmetry, not an internal gauge factor — so BST derives the electroweak algebra ($\mathbb{C}\oplus\mathbb{H}$) with *no internal color group*, where Connes inputs $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ with color gauged. The gap between Connes' complex $M_3(\mathbb{C})$ and BST's real $\mathbb{R}$ is exactly the census result; we are not reproducing Connes' A_F, we are deriving a *different, smaller* internal algebra and explaining why color is external.

Verdict, stated plainly: BST's novelty is *deriving the inputs* — the electroweak algebra (and that color is *not* internal), the generation count, the handedness, and the missing right-handed neutrino that NCG writes in by hand. Connes' spectral action still derives more of the *dynamics* (the Higgs sector, the unification-scale couplings) than we do, and we do not claim otherwise. The two programs are complementary: one derives the discrete skeleton from geometry, the other the continuous dynamics from a spectral action.

7. Honest scope

The reading calculus is a discrete instrument: a fact is readable off I_1 iff it is constant on the reality-type classes $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$. This is *why* the theorem predicts its own negatives — masses, mixing angles, and the CP magnitude are not class functions, so they are provably outside F. They live in a different instrument (Toeplitz/Bergman eigenvalues, the mixing-frame calculus) and are not claimed here. And within that instrument the lepton mass sector needs a *different* mechanism than the quark one: the down-sector Gram/overlap-norm form that derives the quark ratio m_s/m_d = 20 is down-sector-specific (F506) and does *not* transfer to the leptons — whose forced addresses $\{5/2, 3/2, 0\}$ carry a self-shadow singularity at the electron ($\nu = 5/2$) that the quark radial addresses do not. So a lepton-mass claim cannot be read off the quark overlap norm; it is a separate open problem. "Structure, not spectrum" is thus a theorem about the calculus, not a hedge. The one dimensionful input (a Planck-scale ruler) plus the one measured integer n_C are the honest total cost of the discrete internal skeleton — with the ruler forced by dimensional analysis and n_C forced within D_IV by the five-constraint uniqueness (the type-IV Yang–Mills squeeze). And the internal *content* of that integer, for the census, is one trinary question: 1.58 bits decide the weak force.

Appendix A. Cross-domain minimality (the companion result)

This appendix answers "why exactly two invariants generate the skeleton, and no more" by stepping back to all bounded symmetric domains. It is a proof technique for the minimality claim, not part of the one-integer physics lead — Casey's re-centering (Round 17) puts it here deliberately. The sibling domains D_IV^{4,7,8} appear only as separators: foils that fail when a generator is dropped.

The minimal generating set is exactly two: the census (I_1) and the boundary's non-orientability (I_2) — proved both ways. Two separators establish that neither can be dropped (below); and the rank is not a third generator — it is derived from the census. This is the correction of Round 20, and it *strengthens* the theorem: "exactly two" is a sharper claim than "three, all independent."

SEP-3 refuted (the rank is census-derived, not independent). A separator proves an invariant *independent* by exhibiting a domain where dropping it loses a fact. SEP-3 — the attempt to prove the rank independent — returned the opposite: the census *forces* the rank. A quaternionic spinor block caps the rank at 2 (a rank-≥3 domain cannot carry the SM census's $\mathbb{H}$ block; T1443), so once the census has its quaternionic block the rank is pinned to 2, and the three generations (rank + 1) are a census-forced channel, not a free input. So there is no separator making the rank independent — there is a *proof it is derived*. (A search that ends in a no is written refuted, never "closed.")

The ladder (two counts, one per reference class). With the rank derived, the ladder has two rungs, not three:

rung	quantifier (reference class)	independent generators	what is derived
2	∀ bounded symmetric domains D	2: census, orientability	rank (the census's quaternionic block caps it at 2 → 3 generations)

rung	quantifier (reference class)	independent generators	what is derived
1	∀ D in the type-IV family	1: n_C	census + orientability (both functions of n_C)

Going from rung 2 to rung 1 is a *trade*: a narrower reference class (the type-IV family) in exchange for collapsing the two generators to one integer. Rung 1 — the physics lead of the main text — is the most specific and cheapest, and the one that applies to the physical domain. Never "3-of-3": the generating set is two, universally.

Rung 2 → Rung 1 (n_C determines census and orientability). Within D_IV^n the two remaining invariants are both functions of n = n_C: the census through the spinor reality-type (Bott/ABS mod 8) and the color multiplicity (n_C − 2); the orientability through the Shilov parity (mod 2). This is the two-clocks structure of the main-text lead.

The siblings as minimality separators (the family check itself is in the lead, Section 1). The six-row family table of Section 1 already shows *that* the siblings differ; here they do the distinct job of proving *non-droppability*. The two invariants are locked to n_C by two different clocks — the census by Bott/ABS mod 8, the orientability by the Shilov parity mod 2 — so neither can be recovered from the other, and a valid separator must move exactly one of them while holding the rest fixed.

K1744 corrects the separator indices. The census separator is D_IV^7 (a domain whose census differs from the SM's, so dropping I_1 loses a census-keyed fact). The twist separator is the pair D_IV^7 vs D_IV^8 — same real-color family, differing Shilov parity (7 odd / non-orientable, 8 even / orientable) — so a fact that flips between them is the fingerprint of I_2. D_IV^4 was never a valid separator: its census is ($\mathbb{C}$, $\mathbb{H}$, $\mathbb{C}$), not the SM's ($\mathbb{C}$, $\mathbb{H}$, $\mathbb{R}$) — at n = 4 the color space has dimension 2 and SO(2) is *abelian*, so the color block acquires a complex structure and goes $\mathbb{C}$. That is an SO(2)-abelian accident at n = 4, a *different census*, not a same-census/different-orientability foil. (Earlier drafts cited D_IV^4 as the twist separator; it fails because it changes the census too, so it cannot isolate the twist. The valid isolating pair is $D\_IV^{7}/^{8}$.)

Elie's orientability criterion (a type-level answer). A Shilov boundary with connected isotropy is orientable; the chirality-enabling non-orientability requires a *disconnected* isotropy (a $\mathbb{Z}_2$ component). Among the irreducible bounded symmetric domains, only types III and IV can carry that $\mathbb{Z}_2$ twist — so most domains cannot host a chiral, Majorana world at all; the requirement is not generic. Positive control: $D\_III^2 \cong D\_IV^3$ (the exceptional isomorphism $Sp(4,\mathbb{R}) \cong Spin(3,2)$) — the one domain in both admissible families; the census and orientability agree across the two descriptions, as they must.

Bit-ceiling. With the rank *derived* from the census (no independent bits), the two generators cost census capacity ≈ 4.75 ($\log_2 3^3$) + orientability 1 ≈ under 6 bits for the whole discrete internal structure across the reference class. *Within* D_IV the census content collapses to the single trinary spinor question, 1.58 bits (+ 1 orientability) — the figure the main text leads with.

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v0.7 — Round 20: SEP-3 REFUTED → minimal generating set = exactly two invariants (census + non-orientability); the rank is derived from the census (quaternionic block caps rank at 2), not a third generator. "Exactly two" beats "three, all independent." Kept distinct: the invariant generating count (two, Appendix A) vs the physical selection conditions (three, lead, Strong-Uniqueness). Appendix A ladder collapsed 3/2/1 → 2/1; bit-ceiling under 6. Status scrubs: removed stale "four conditions" and "unique quaternionic." Prior — v0.6 Round 19/§644: real-color rides on non-orientability (four→three selection conditions); N_c=3 scoped as a reading of n_C. v0.6 — Round 18: lead corrected + locked to Strong-Uniqueness (K1697). "Unique quaternionic" was false (4,5,11,13 quaternionic); the honest form is smallest-of-conditions with the measured N_c=3 as tiebreaker among {5,11,13,...}. Six-row family table moved into the lead (cells derived from H_F(n)); bits labeled (spinor 1.58, family census 2.00). SEP-e split kept (End_K = $\mathbb{C}\oplus\mathbb{H}\oplus\mathbb{R}$; bare End_$\mathbb{R}(V_{12}) = M_3(\mathbb{R})$ — not blanket-replaced). Prior: v0.5 — Round 17 re-center (Casey: "why are we even talking about D_IV^4?"). One-integer / one-trinary-question is the lead; three-invariant cross-domain minimality + all siblings (D_IV^{4}/^{7}/^{8}) demoted to Appendix A (companion / proof-technique). K1744 folded: ordered-triple multiplicity-free census (hygiene, not load-bearing); census content = 1.58 bits + family census 2.00 bits (not 4.75 capacity); weak force = quaternionic spinor at n ≡ 3, 4, 5 (mod 8) — n_C=5 is the smallest residue that also passes the other selection conditions (NOT "unique quaternionic," which was false); twist separator D_IV^7 vs D_IV^8; D_IV^4 invalid (census ($\mathbb{C}$,$\mathbb{H}$,$\mathbb{C}$), SO(2)-abelian accident); algebra $\mathbb{C}\oplus\mathbb{H}\oplus\mathbb{R}$. Sources: F1060–F1067 (arc); K1724 (census), T1949/T2522 (chirality/twist), T2525 (generations = rank+1), T1443 (type IV rank-2), T2551 (Connes), T265 (Coleman–Mandula), K1743 (bare-End correction), K1744 (separator/bits correction), Elie's anomaly + orientability criteria. Ships on Keeper-PASS + Cal-vet + both voices + Casey GO. Nothing pushed; CP existence-only.

— Lyra, Thursday 2026-08-20 (date-verified).